

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The network protocol analyzer logs indicate that port 53 is unreachable when attempting to access yummirecipesforme.com. Two requests were sent to the DNS server with UDP protocol. Based on these results, an ICMP Destination Port Unreachable returned an error message, consisting in: "ICMP 203.0.113.2 udp port 53 unreachable length 254". That port is used for a Domain Name System server to resolve domain name in IP addresses. In this way, the most likely issue is that no DNS service was listening on UDP port 53 on the DNS server.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred at 13:24 when multiple clients tried to access www.yummyrecipesforme.com and received an error message: "Destination port unreachable". The IT department opened a Network Protocol Analyzer, called "tcpdump" and captured two requests sent to port 53, used by the DNS servers to resolve domain names into IP addresses. The suspected cause of the incident is that no DNS service was listening on port 53. In this way, a possible DoS attack may have occurred or a member of IT department configured firewall to block traffic to 53 port.